

Trade Diary Application: Core Functionality Specification

This document provides a comprehensive, centralized breakdown of all core functionalities discovered across the Trade Diary application modules. Use this as a functional specification blueprint to build out the feature set for your own journal platform.

1. Dashboard Module

1.1 Primary Metric Tracking

- **Performance Metrics:** Captures and shows net Profit & Loss (P&L), win rate percentage, and total trade counts compared to a previous month's baseline.
- **Risk Evaluation:** Displays calculated average Risk-to-Reward (R:R) ratios across positions.
- **Market Tickers:** Displays a real-time status bar for broad market indices (e.g., Nifty FMCG, Nifty Healthcare, Nifty IT, Nifty Media, Nifty Metal, Nifty Pharma) directly at the top of the interface.

1.2 Psychological Tracking

- **Confidence Index:** Converts qualitative emotional input into a quantitative visual scale to measure whether a trader is following their plans or trading out of fear over a 30-day period.

1.3 Core Analytical Charts

- **Cumulative P&L Chart:** Displays a smooth-line gradient area graph tracking compounding equity growth or drawdown over specific time periods (Days, Weeks, Months).
- **Top Trades Ledger:** Siphons out and ranks the single highest-grossing trades of the month, displaying the absolute profit, percentage gained, exact entry/exit prices, and execution dates.

- **Win/Loss Distribution:** Renders a dynamic donut chart showing the exact numerical split between winning trades and losing trades alongside the absolute win rate percentage.
 - **Strategy vs P&L Chart:** Provides a comparative vertical bar chart plotting total profitability or loss against specific named trading strategies (e.g., Breakout, News-based, 9&15 Ema, Pullback).
 - **Most Common Mistakes Tracker:** Aggregates manual psychological tags to show a progress-bar ranking of behavior issues alongside the total number of occurrences and the financial loss tied to those specific mistakes.
 - **Daily P&L Bar Chart:** Plots daily net outcomes chronologically on a bar chart to easily identify daily win/loss streaks and volumetric consistency.
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2. Trade Execution & History Module

2.1 Manual Data Entry ("Add New Trade" - General Tab)

- **Trade Duration Setup:** Selectable toggle options to categorize trades as either Intraday or Swing positions.
- **Trade Metadata Fields:** Collects core setup information including market type (e.g., Indian), asset Symbol/Ticker search entry, and Entry Date calendar picker.
- **Execution Metrics Entry:** Standard numeric input fields for entry price, quantity, exit price, stop loss, and target.
- **Auto-Calculated Output Fields:** System-generated readout blocks that display computed metrics like Total Amount, P&L Amount, and P&L Percentage based on manual price entries.
- **Directional Mapping:** Clickable toggle buttons to lock in position direction as either a Long or Short trade.
- **Categorization Tags:** Built-in dropdown fields to link the execution to a pre-saved "Strategy" and assign an "Outcome Summary" tier.

2.2 Qualitative Logging ("Add New Trade" - Psychology Tab)

- **Confidence & Satisfaction Scales:** Independent horizontal sliders mapping Entry Confidence Levels (1-10) and Satisfaction Ratings (1-10) from low/unsatisfied to high/satisfied.
- **Emotional State Tracker:** Dropdown menu allowing the user to select their predominant emotional state felt during active trade execution.
- **Mistake Checkbox Array:** A multi-select matrix containing common behavioral trading pitfalls (e.g., Overtrading, Risked Too Much, Exited Too Late, Ignored

Signals, Ignored Stop Loss, Greed, Revenge Trading, Exited Too Early, FOMO Entry, No Clear Plan, No Mistakes).

- **Qualitative Lesson Logs:** An open-ended text box dedicated to typing out structured post-trade reflections or lessons learned.
- **Technical Documentation Fields:** An open-ended text block for a detailed structural “Trade Analysis” narrative alongside an image upload area for attaching chart screenshots.

2.3 Master Ledger History

- **Data Density Grid:** Displays a row-by-row matrix of all recorded positions tracking execution Date, Symbol, Direction, Entry/Exit levels, Net P&L (with underlying return percentage), R:R, Strategy tag, and Outcome badge.
- **Action Shortcuts:** Contextual inline row icons to instantly Edit or Delete specific trade history lines.
- **Global Ledger Utility Controls:** Top-level buttons allowing users to Sort rows, Apply multi-variable Filters, Sync data feeds, or Bulk Import raw records.

2.4 Broker Integration Portal

- **Direct API Connection Hub:** A dedicated onboarding modal displaying regional broker options (e.g., Dhan, Angel One, Zerodha, Upstox, Fyers) to authenticate account handshakes for fully automated transaction logging.
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3. Trading Strategies Module

3.1 Strategic Performance Cards

- **Categorized Profit Calculations:** Tracks and breaks down cumulative capital growth or decay separately under distinct strategic tags (e.g., Breakout, Pullback, News-based, 9&15 Ema).
 - **Setup Efficiency Metrics:** Computes and prints specific advanced mathematical performance formulas such as the Profit Factor and Risk per Trade for each independent strategy bucket.
 - **Strategic Success Meters:** Displays individual win-rate percentage bars color-coded by performance thresholds to isolate highly optimized strategies from broken setups.
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4. Calendar Module

4.1 Date-Based Visual Journaling

- **Daily Outcome Overviews:** Displays a standard monthly grid view showing total daily net profits/losses and total trade execution volume directly inside the individual date blocks.
 - **Color-Coded Equity Mapping:** Uses distinct conditional background colors (e.g., green for profit, red for loss, dark gray for no trades) to give a visual overview of a trader's monthly performance.
 - **Single-Day Deep-Dive Modal:** Clicking an active calendar cell activates a "Trading Details" pop-up. This module summarizes that specific day's cumulative P&L, calculated Average R:R, total trade volume, net win rate, and renders an isolated sub-ledger listing detailed execution statistics (Symbol, Side, Size, Entry, Exit, P&L) for just that day's positions.
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5. Trading Challenge Module

5.1 Milestone Goal Configuration

- **Target Capital Parameters:** Numeric input form fields to establish an account baseline (Starting Capital) and a strict target goal milestone (Target Capital).
- **Timeframe Controls:** A dropdown system to bound the duration of the operational challenge (e.g., 1 Week).
- **Risk Parameters:** Dedicated boundary entry box to set up hard rules like a maximum allowable risk percentage limit per individual execution (Max Risk Per Trade %).
- **Operational Filters:** A toggle option to exclude non-trading market holidays or weekend breaks from the challenge's active timeline logic.

5.2 Progress Dashboards

- **Goal Completion Ring:** A prominent radial percentage progress wheel displaying how close the user's current account balance is to hitting their set milestone target.
 - **Dynamic Pace Calculators:** System computing cards that track Days Remaining alongside a real-time, variable calculation displaying the exact daily profit target rate needed to cross the finish line on time.
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6. Advanced Analytics & Reports Module

6.1 Quantitative Capital Audits

- **Capital Usage Profile:** Tracks maximum, minimum, and average cash exposure thresholds alongside tracking exact net performance metrics earned at peak capital exposure levels.
 - **Sizing & Quantity Audits:** Correlates profit efficiency against share/lot sizing, displaying performance variations captured at maximum vs. minimum trading quantities.
 - **Temporal Performance Mapping:** Generates detailed analytical matrices isolating historical Win Rates, Average Trade Volume, Max Trades in a Day, Days with Only 1 Trade, and Overtrading Days broken down across the active days of the week (Monday through Friday).
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7. Mistakes & Rules Management Module

7.1 Behavioral Analytics

- **Mistake Density Meters:** Tracks global mistake counts over defined rolling time blocks alongside percentage trend tracking lines.
- **Error Distribution Charts:** Renders a vertical bar chart measuring the exact frequency of specific psychological mistakes (e.g., No Mistakes, Risked Too Much, Greed, Overtrading, Revenge Trading, Exited Too Early, FOMO Entry).
- **Behavioral Frequency Heatmap:** Maps out a weekly grid layout that maps occurrences into clear color density blocks to discover seasonal trends or behavioral slip-ups over time.

7.2 Disciplinary Rule Creation & Tracking

- **Rule Configuration Engine:** A modal to define operational boundaries containing fields for Rule Name, Execution Phase Categories (e.g., Entry, Management, Exit), and a detailed plain-text Description box.
 - **Compliance Matrix Displays:** Dynamically tracks rule compliance by separating rules into two comparative visualization tables: "Top 5 Most Followed Rules" and "Least Used Rules" based on manual trade logs.
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8. AI Summarizer Module

8.1 Configuration Controls

- **Timeline Boundary Selection:** Button controls to isolate historical ledger data into distinct analytical blocks (Last 30 Days, Last 60 Days, Last 90 Days, or Custom Range).
- **Context Scopes:** Regional dropdown selector to isolate calculation variables based on specific geographic markets.

8.2 High-Level Summary Card Grid

- **Efficacy & Volume Calculations:** Extracts macro metrics over the selected timeline: Total Trades, Win Rate %, Total Net P&L, Financial Expectancy, Avg Win, Avg Loss, Confidence score, and Avg R:R.

8.3 AI Generative Synthesis Reports